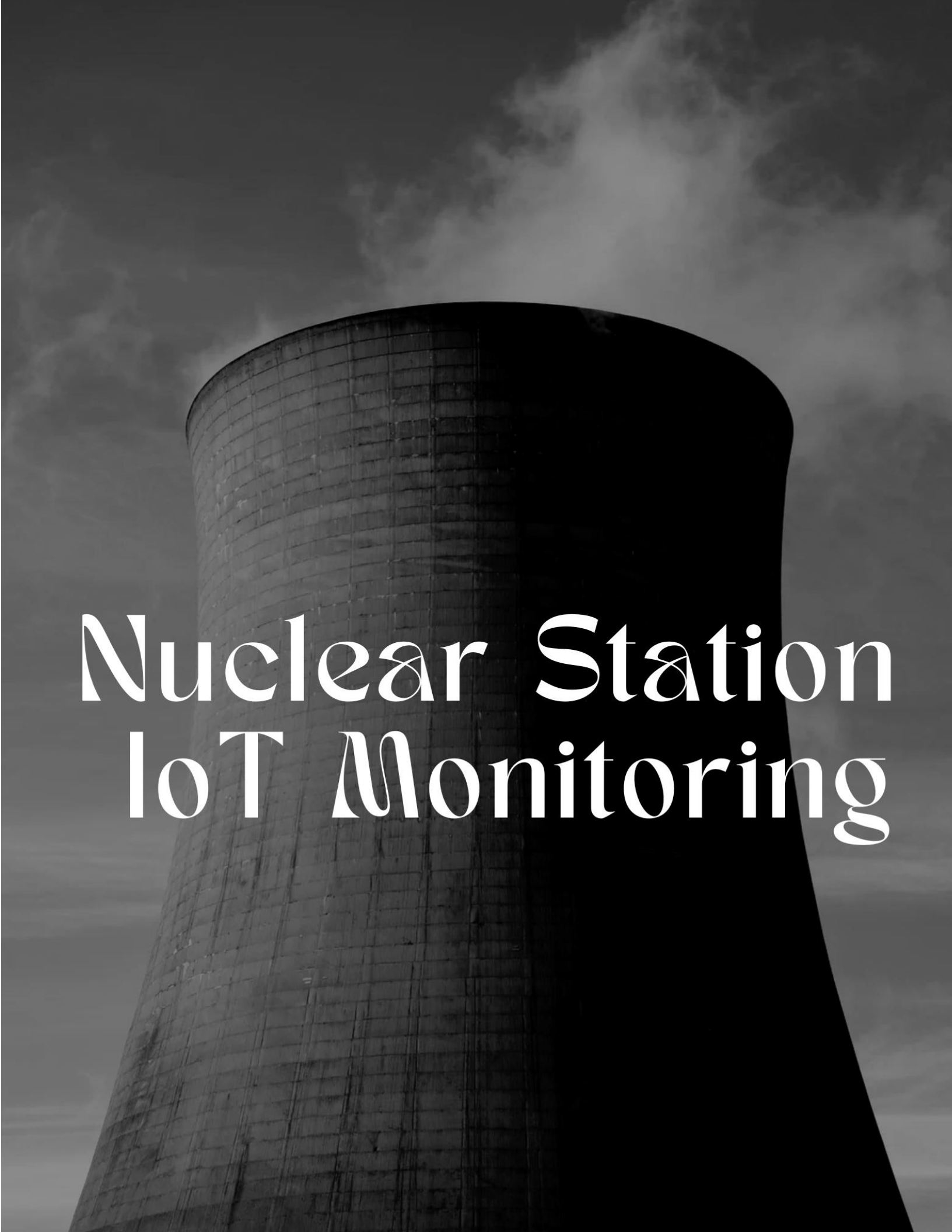

Nuclear Station IoT Monitoring

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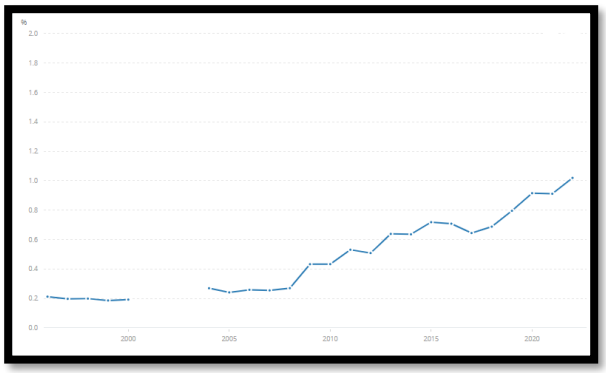
1.1 Egypt Grand Challenge/s

Improve the scientific and technological environment for all.

The challenge of improving the scientific and technological environment is a critical issue that has been tackling Egypt's overall development. Addressing this challenge is crucial for its economic growth, high-quality job creation, education, innovation, and skills development. It also ensures that Egypt develops along the lines of energy production, medical advancements, environmental sustainability, and enhancement of national security through enhanced technological independence and cybersecurity. Moreover, it increases Egypt's international standing through international partnerships and cultural impact. In short, this challenge, when addressed, can greatly enhance the quality of life in Egypt and put the country in a leadership position in innovation and research.

Figure 1.1:
Research and development expenditure (% of GDP) - Egypt,



Insufficient funding for research and development (R&D) is one of the challenges facing the country. Egypt's government allocates a relatively small portion of its Gross Domestic Product (GDP) to R&D compared to many developed nations. This lack of funding limits the ability of scientists and researchers to conduct groundbreaking studies and develop new technologies. In December 2022, Egypt came in **35th** place in terms of public spending on research and development R&D among 86 countries worldwide, spending only **1.02%** of its GDP on R&D which is very low compared to leading countries like 3.46% of the United States and 2.27% of European Union. (Media Center, 2022) (Research and development expenditure (% of GDP), 2024).

There is also limited Collaboration Between Academia and Industry, there exists a disconnect that blocks the process of knowledge and technology transfer. The universities are mostly unable to commercialize research outputs, whereas the private sector is probably unable or not keen to invest in research collaboration. In this way, it bottlenecks innovation and economic growth. (Darwish, 2022)

Regarding Infrastructure and Access to Technology in Egypt, there is still much that needs to be done, especially in rural areas, considering the Egyptian country has done quite a lot with bettering infrastructure and increasing access to technologies. Their efforts to expand broadband access, improve the reliability of power, and upgrade facilities for laboratories are necessary for advancing scientific and technological landscapes within the country. Investment and focus need to be continued until the benefits of technology can be accessed by all citizens. However, this sector is one of the most promising in Egypt.

While Egypt's startup scene has, in fact, grown, especially in the tech space, there remain plenty of challenges: a lack of access to venture capital, much bureaucratic red tape, and limited opportunities for mentorship. Indeed, from government support and public-private partnerships, financial support, and incubation centers to tech hubs for startups, especially those related to technology innovation and scientific development, could provide possible catalysts for an improved entrepreneurial culture harvesting science and technology.

In conclusion, both technology and science are indispensable in this developing phase. By leveraging technological innovations informed by scientific research, we can work towards higher quality type of work in all sectors, and build a more sustainable future for generations to come.

1.2 Problem to be solved.

Aligning with improving technological environment, improving traditional methods used in monitoring energy production and the parameters surrounding it like safety and efficiency stands out. While human operators have historically played a crucial role in monitoring energy production stations, their limitations become increasingly apparent when compared to the capabilities of sensors and IoT (Internet of Things) technologies. These limitations include multiple disciplines.

Humans cannot see and simultaneously observe all relevant multiple parameters continuously, often leading to delays in responding to a critical event. The fact is that human observation can be subjective, with the propensity for errors, especially under conditions of stress or when data is complex. These kinds of mistakes could further lead to unsafe conditions and operational inefficiency. On the other hand, sensors can deliver real-time data for the immediate detection of anomalies and rapid responses, in addition to very precise objective measurement. Automation via sensors and IoT would be able to reduce human error and thereby improve total plant efficiency. A person tasked with monitoring something for long periods is often prone to fatigue and lesser alertness, hence bringing down the accuracy of the observation. Sensors offer consistent and reliable data collection, irrespective of human factors. Furthermore, human capabilities are limited in analyzing massive volumes of data and making informed decisions in the least possible time. IoT devices integrated with modern analysis techniques will be able to process data in real time, detect trends, and make recommendations towards achieving optimal operations. Financially, there are a lot of costs due to late maintenance and energy wasting that come up with traditional monitoring. A study conducted by Deloitte reveals that predictive maintenance can curtail maintenance costs by as much as **30%** and a report by the World Economic Forum elucidates that predictive maintenance can diminish maintenance planning time by **20-50%**, augment equipment availability by **10-20%**, reduce overall maintenance costs by **5-10%**, and indicates that Industrial Internet of Things (IIoT) can yield energy savings of up to **20%** in manufacturing environments. (Green, 2023).

Nuclear plants are particularly burdened to monitor parts of the generators because these are high-risk facilities. This is due to the huge consequences from failures or accidents that might occur in such plants; therefore, monitoring of vital parameters is required with a great degree of accuracy and at the right time. For instance, the World Nuclear Association reports that human error is one of the most common contributors to nuclear accidents. Further, the complex, interconnected systems at nuclear power stations may be beyond the capability of human operators to monitor all the relevant parameters and to recognize the unfolding problems. According to research conducted in the University of Texas studying the likelihood of errors being made by staff members responsible for dangerous technologies and the factors that can increase the risk, almost **80%** of incidents recorded at the ten nuclear energy centers that have been examined were the result of either a worker's error or the use of poorly designed procedures. (Bohigas & De Fortuny, The incidence of human error in the handling of nuclear weapons, 2024)

If the problem is solved:

1. **Reduces Human Error:** Automation powered by sensors, IoT will definitely mean a decrease in human error especially under stressful or complicated situation.
2. **Higher Plant Efficiency:** Predictive maintain from real-time data to make the plant operations more efficient.
3. **Safety:** Any anomalies are quickly identified, prompting immediate response, and all but eliminating the risk of accidents.
4. **Increased Equipment Availability:** extends the lifespan of machinery while minimizing potential downtime besides rendering mission-critical machinery more available for use.
5. **Cost Savings:** With Predictive maintenance; you can reduce maintenance cost, save energy by going for predictive measures and have less unplanned shutdowns.
6. **Higher Accuracy in Monitoring:** Consistent and objective data collection improves monitoring accuracy, avoiding fatigue-related errors.
7. **Better and faster decision-making**: Combining IoT with data analytics facilitates real-time analysis, trend identification, and actionable insight.

If the problem is <u>not</u> solved:

1. **Greater Chances of Accidents:** The reliance on human control and slow identification of deviations from the norm can create unsafe environments and accidents.
2. **Reduced Plant Performance:** Excessive dependence on manpower for monitoring can slow down response time and result in ineffective processes.
3. **Rising maintenance Expenditures:** Unbalanced assessments of maintenance activities relate to considerable losses because of visuals and breakdown of equipment and systems.
4. **Extended Downtime:** Downtime that is not scheduled in advance due to failure to detect anomalies in good time or the equipment breaking down may pose losses in power generation and increase the costs of running the operations.
5. **More Human Fatigue:** Keeping staff on their toes all the time will lead to tiredness, poor vigilance resulting into poor changes detection.
6. **Inferior Data Evaluation:** Averages in human ability to work with extensive information and a database unnecessarily leads to lost chances of improvement.

1.3 Research

<u>Topics Related to the Problem:</u>

a. **Lack of full-time Automated Monitoring in the industrial sector**
   Egypt hasn't depended on automation in industry with a complete form. Full-time automatic mentoring systems aren't widely used in Egypt's factories. Manufacturers are unaware of critical changes such as temperature deviation, gas leaking, or equipment malfunction. This lack of critical feedback increases the risk of accidents. In addition, the lack of automatic monitoring systems plays a role in reducing production quality. The inability to track, collect, and analyze data easily contributes to not taking actions that can optimize the production process. High and medium high tech, which includes pharmaceuticals, electronics, automobiles, and aerospace construction, currently account for 14% of manufacturing value added (MVA), which is significantly lower than both the average of 37.3% for nations in an industrialization stage comparable to Egypt and the overall global trend. (Are Egypt's manufacturing industries embracing the shift to industrial automation?, 2022)
b. **Monitoring and Environmental impacts**
   Undetected deficiencies such as gas leaking, UV leaking, or even temperature deviation may cause environmental impacts such as contaminating water, air, and soil. In addition, inaccurate monitoring can lead to insufficient use of resources, increasing the consumption of energy and materials.
c. **Low accuracy and delays in the manual (labor-depended) Monitoring**
   In Egypt's industrial sector context, factors still rely on manual monitoring. Manual monitoring isn't sufficient due to the lack of real-time data updates and delays in collecting and analyzing data. Huge crises happen due to the delay in human tracking. Furthermore, manual monitoring has a high rate of human error which can contribute to extreme errors in safety-critical industries such as energy production in nuclear stations or pharmaceutical factories.
d. **Lack of Mentoring in nuclear power stations**
   In meticulous manufactures, mentoring should be done with the highest levels of accuracy. Many operations contribute to producing energy and aren't mentored well, starting from the nuclear fusion reactions to delivering energy into usage. Lack of mentoring systems in nuclear power stations leads to

unawareness of important factors that should be controlled well or it will lead to crises like what happened in Chornobyl due to lack of real-time monitoring resulted in increasing temperature in the 4th reactor till the explosion.
The temperature of the nuclear core reactor can deviate instantly with no pre-signs; so, it should be tracked because any deviations in it may indicate a sign of an explosion. The rate of flow of the water used to cool the core of the reactor should be measured as well to ensure the core will be cooled well. In nuclear power plants, UV radiations, invisible light, often escape from the reactor, causing severe side effects to the workers.

Topics Related to the Solution:

a. Automated Mentoring system

Automated mentoring systems rather than manual ones are typical solution factories aiming to implement more widely to increase the awareness of the operating conditions. The advantages of these systems lie in two pros:

1. Collecting and analyzing data automatically, which increases the rate of data processing and even speeds up the actions taken by workers.
2. Dynamic data visualization in real-time, which enables the manufacturers to access data at any time to get a full view of the manufacturing process.

b. Mentoring a Nuclear Power Plant

Nuclear power plants are considered one of the most safety-intensive sectors due to the extreme operations done in the reactors. There are 3 factors mainly responsible for producing energy in its entirety. Mentoring those factors indicates how the process of energy production is going.

1. The temperature of the nuclear core reactor
2. The flow rate of the cooling water
3. Radiated UV

These factors affect each other; so, they should be measured and relate their values to predict the efficiency of production.

A nuclear reactor's core temperature must be closely regulated to prevent overheating and take action in the event of a potential threat.

Moreover, reactor cooling water is essential to sustaining the reactor’s temperature; so, any abnormal rates of the fluid water may lead to temperature deviations. UV radiation levels indicate possible radiation leaks or structural integrity issues in containment systems. Monitoring of these levels is essential for maintaining the safety of the operations.

1.4 Other solutions already tried.

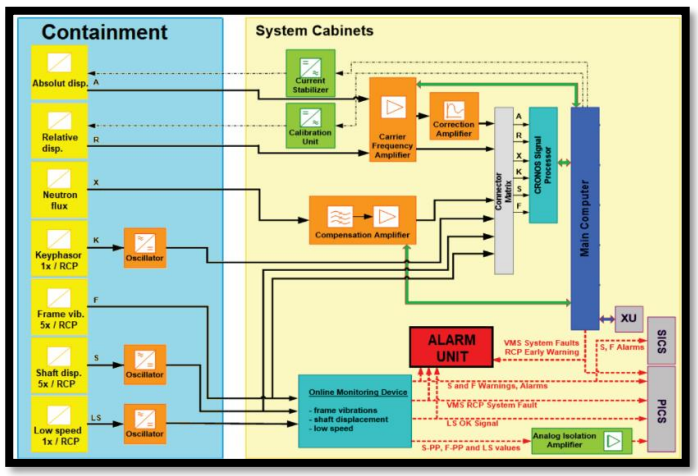
1-The Olkiluoto NPP

The Olkiluoto nuclear power plant is a 3-unit nuclear power plant, and alongside Loviisa plant, makes the only two powerplants in Finland. The first two units of the Olkiluoto (OL for short) were built back in the seventies, while the third had its construction begin in 2005, and continued until 2021. The addition of the 3rd unit was crucial step-up in Finland's energy production, as it raised the share of energy from the Olkiluoto NPP to **31%** of the country's total production, the 3rd unit being responsible for **42%** of that (Teollisuuden Voima Oyj, 2024).

Figure 1.2:
The olkiluto Nuclear power plant unit 3

One of the most important features of the new unit added, was its monitoring system. It consists of multiple system, each measuring a specific aspect of the system. The most important were the JYG, JYM, and JYE, monitoring the first circuit, second circuit, and the rotating machinery respectively. All these systems function automatically, and send information of the state of the reactor to the technical team, which analyze the data, and inform the engineering department in case any unwanted deviation appears.

Figure 1.3:
System layout of JYG vibration monitoring system



In the primary circuit, the JYG mainly measure the vibration of the RPV (Reactor Pressure Vessels) and RCP (Reactor Coolant Pumps). The vibration of the RPV is acquired using rather simple mechanical vibrations sensors at the outer surface of the vessels. In the RCP, The shaft's (the rotating component) low speeds are measured by proximity probe. Otherwise, the vibration is measured by velocity sensors that measure absolute frame (structural component that houses the machinery) vibration and relative shaft displacements between shaft and frame. The

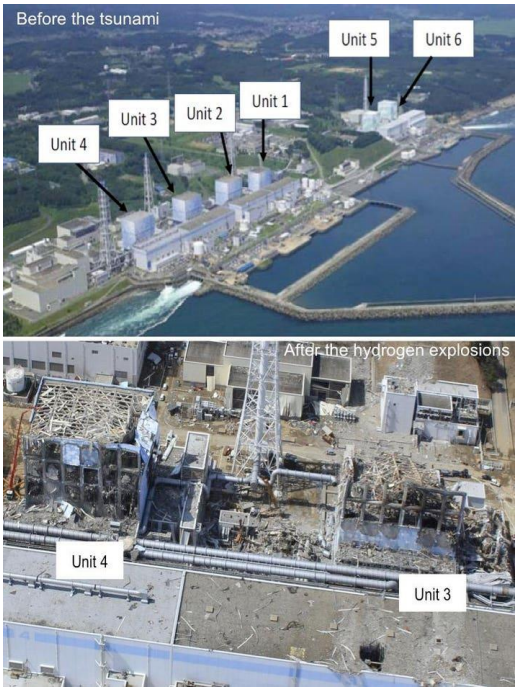
RCP monitoring is crucial, and any problem in it must be identified as quickly as possible, as any problem in it can cause a serious production decline (Rostedt & Koivunen, 2021).

The problem with this method is that while it provides the most accurate results, it can be a little costy, and for a country that tries to get into the nuclear field, might need to check cheaper options.

2-Fukushima Daiichi Nuclear Power Plant

The Fukushima Daiichi power plant was first commissioned in Japan in 1971, consisting of six main units of boiling water reactors. Those reactors worked together, making Fukushima one of the biggest 15 reactors in the world, with a massive production of 4.7 GWe (Saad, 2015). It was also the very first NPP to be planned and built with Japanese companies (General Electric and Tokyo Electric Power Company). All of these made the Fukushima Daiichi power plant one of the most important in Japan.

Figure 1.4:
The Fukushima Daiichi, before and after



In March 2011, a massive 9.0 earthquake occurred in the Pacific Ocean that lasted for a whole 6 minutes and caused a huge tsunami. Unfortunately, the tsunami hit the reactor and disabled the power supply and cooling systems of the reactors. If those damages were discovered earlier, they could have been fixed. However, the system failed to detect the severe damage, causing a delay in response, which eventually led to the Fukushima nuclear accident.

On the INES (International Nuclear Event Scale), the accident was rated seven, the maximum possible severity. What happened was a partial meltdown of the fuel rods, causing an explosion in the outer containments and the release of radiation into the air. Though nobody directly died, more than 2000 died of radiation-caused diseases (World Nuclear Association, 2024). The lack of an accurate and adequate monitoring system made the Fukushima power plant, which was one of the most valuable energy resources in Japan, to be abandoned and the area to be evacuate.

2.1 Solution and Design Requirements

Solution Requirements:

To address the problem of energy production monitoring at nuclear plants the solution should be using IoT sensors for an integrated, real-time monitoring system. The goal is to reduce human error, enhance safety, and optimize energy production efficiency. To offer these advantages the system should focus on trends in the critical parameters:

1- Temperature of the core reactor
2- Cooling water flow rate
3- UV radiations.

The solution should consist of **real time monitoring**, processing the data immediately, to provide **reliable detection** and **data analysis**. In addition, the detection and analyzing process should be **automated**, reducing reliance on human labor and approaching accuracy. Finally, our system should reassure the safety of the nuclear power plant to avoid the potential crisis to happen due to system deficiencies.

Design requirements:

To highlight the success of the prototype, the design requirements represented in the two following points should be met:

1) **UVC light presence:** UV light should be detected, determining the presence or absence of UVC type, by analyzing the detected UV wavelengths, as it's the most common type in nuclear plants, ensuring the plant's safety.
2) **Real-time monitoring**: The trends in the traced variables should be updated on the dashboard within a maximum of 10 seconds continuously to enhance data analytics and increase the efficiency of the nuclear plant.
3) **Emergency Alert:** An emergency alert should be launched when the core temperature approaches X °C lower than the forging temperature.

2.2 Selection of Solution

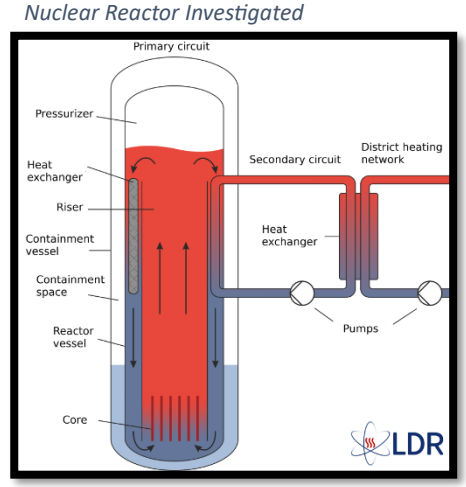
In this respect, a whole solution package that automates the monitoring in such a nuclear power station can reduce the huge risk of accidents occurring due to insufficient monitoring. This will be realized through advances in digital sensors and IoT technology. In this manner, the system will carry out an assignment of **continuous monitoring** of vital parameters (Nuclear Core Temperature, Coolant Water Flow Rate, Radiation Leakage), **process data gathered**, and **give insights** in user-friendly terms for top technical and nontechnical specialists to make decisions on safety, maintenance, or other common themes like efficiency of operation.

Monitored Factors:

1. Nuclear Core Temperature:

The core temperature fluctuates in response to changes in reaction speed, which is controlled by the insertion or withdrawal of control rods. As reaction rates increase, control rods are inserted to reduce particle collisions, thereby limiting heat generation. **Real-time monitoring** of temperature variations enables precise and timely adjustments to control rods and reaction speed. **Advanced data processing algorithms** is applied on the temperature recordings, predicting critical temperature threshold crossings, like the **forging temperature** (the temperature at which a metal becomes substantially softer, but is lower than the melting temperature, such that it can be reshaped by forging or by applying small forces). This predictive capability allows for proactive maintenance strategies that can prolong the life of the core material and minimize the risk of catastrophic failures.

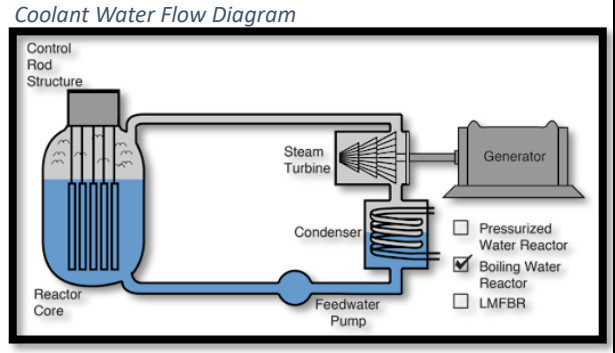
Figure 2.1:
Nuclear Reactor Investigated



2. Coolant Water Flow Rate:

In nuclear reactors, water is used to draw off the heating produced by nuclear reactions that turn later to steam in order to rotate turbines. The rate of flow of the coolant water should be monitored to avoid overheating due to poor dissipation of heat. Measurement of the flow rate of coolant water according to the core temperature is an

Figure 2.2:
Coolant Water Flow Diagram



important **correlation** in determining possible malfunction-for example, of pumps-or reduction in the flow of water that may affect safety conditions of the reactor and energy production.

3. Radiation Leakage:

Since nuclear power has hazards associated with it, any leakage of radiation has to be closely monitored. Any seepage through containment will have far-reaching consequences on the environment and health. The nuclear radiation has variable penetrating power according to type of particles: alpha, beta, gamma, etc. However different types of insulating materials are used to prevent the radiation effect on surroundings, some leakage from the insulating material is accompanied due to factors like aging, radiation damage, thermal stress, installation defects and others. On accidents like core melting a critical indicator is radiation leakage. The automated monitoring system will be able to provide very **early detection of the leak in radiation**, thus responding and mitigating on time to reduce exposure and contamination.

Data Storage and visualization:

For the monitoring process to be meaningful, the data of sensors readings should be collected, stored overtime, then processed and visualized.

Data Storage:

Databases are classified according to storage location into local database and cloud database. A local database store data on physical servers or hardware located on-site of its collection. Cloud databases are hosted and managed in a cloud computing environment and they are accessible over the internet.

Regarding data storage of sensors readings, a **Cloud database** is used where it offers more Scalability, Flexibility and Real-time data processing where:

Figure 2.3:
Cloud Database

Scalability: Easily handles large data volumes and scales with increased sensors or users and avoids the need for upfront hardware investments.

Accessibility: Data can be accessed and managed remotely from any location with an internet connection, so data and insights are easily shared with system users across different locations.

Reliability: Cloud providers offer high uptime, redundancy, and disaster recovery mechanisms where data is replicated and backed-up across multiple servers or data centers to be easily recovered.

Integration: Simplifies integration with other cloud-based tools for visualization and analytics.

Maintenance: Cloud providers handle database updates, security and maintenance tasks.

Elasticity: Dynamically adjusts storage and processing resources based on current demand.

Data Visualization:

The collected and stored readings are visualized in **user-friendly dashboard** to change quantitative data into graphs, charts and gauges that give insightful qualitative meanings and visualize changes in monitored variables in a way that can be understood by **technical and non-technical users** indicating **trends** in data behavior and threshold limiting and crossing **threats**.

Finally, the proposed solution of automatic monitoring system will come with great advantages to nuclear power plants. It allows the operator to act in advance of impending safety threats by carrying out continuous, real-time monitoring of all vital parameters-core temperature and coolant water flow rate to radiation leakage. Data-driven insights enable predictive strategies for maintenance, hence optimizing operational efficiency and reducing downtime. User-friendly visualization dashboard provides clear, actionable information for experts and non-experts alike to make decisions. Besides, by ensuring strict adherence to the standard of radiation safety, the system is increasing environmental sustainability. In general, with the implementation of this advanced monitoring solution for nuclear power stations, the safety, reliability, and overall performance of nuclear power plants are brought to a completely new level and align with the **7th** and **11th Sustainable Development Goals (SDGs)** and align with **Egypt vision 2030** of building balanced economy based on technology (THE 17 GOALS, 2024)

Figure 2.4:
7th and 11th (SDGs)

2.3 Selection of Prototype

In order to test the solution and design requirements and meet the proposed solution, a testable prototype is designed and constructed.

The prototype has two disciplines: a system simulating changes in variables that occur in the nuclear power station (temperature, water flow, and radiation), and the monitoring system that monitors, stores and visualizes the variables changing in the simulation system.

Simulation System:

The monitored factors are simulated in the prototype as following:

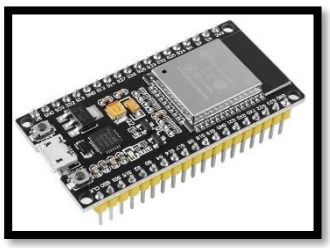
1.**Coolant Water Flow**: In nuclear power stations, coolant water flows form a near source to core under the effect of water pump, so in the prototype water is stored in a 6L container to work as a water source and pumped using a 12V water pump flowing in hose.

2.**Nuclear Core and Radiation Leakage**: Due to complexity, cost, safety, and many conditions, executing nuclear reaction in the prototype is infeasible and unattainable. The factor we consider in the nuclear core is its heat energy used to change water received from pump into steam and its effect on the core material, so whatever the source of this heat is, what really matters is reaching high temperatures that can boil water and change it into steam. In this scope a gas heater is used as a heat source to heat an aluminum container representing the core. Finally, the high energy leaks in nuclear station of gamma, alpha, beta particles are simulated using lower energy type of radiation which is UV radiation using a UV lamp.

Monitoring System:

Monitoring system is composed of **sensors** that keep track of the critical factors of the nuclear power station simulated in the prototype, integrated with a **microcontroller** that collects sensors readings and using **Internet of things (IoT)** starts to send them on servers where they are **stored** and **visualized**.

Figure 2.5:
ESP32



From a wide range of microcontrollers, **ESP32** is considered the most suitable microcontroller for the monitoring system compared with other microcontrollers like Arduino and

Raspberry-Pi for multiple reasons the most important of whom are:

- High processing power and memory (Dual-core 32-bit processor with clock speeds up to 240 MHz) in addition to many GPIO pins with different features and functions.
- Built-in Wi-Fi and Bluetooth capabilities making it ideal for IoT and wireless communication.
- Supports many programming environments like (Arduino IDE, MicroPython, …) making it simple and flexible to be used.

Arduino boards lack built in Wi-Fi or Bluetooth and need external modules while Raspberry-Pi, however providing higher processing power and more features, is more complex and overprice the prototype with many additional features that are not needed, so ESP32 provides functionality and flexibility in a reasonable price matching its demand in the context of prototype.

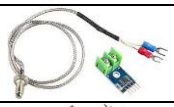
The ESP32 is powered using two **3.7V batteries**, and collects sensors readings of **"YF-S401"**, **"Thermocouple k-type MAX-6675"** and **"GUVA-S12SD"** that measure water flow rate, aluminum container temperature, and UV light concentration respectively.

The collected readings are then sent using internet connection and **wirelessly** to the server of **"Adafruit-IO website"** using message query telemetry transport **(MQTT)** IoT protocol where ESP32 works as a **publisher** and the server of the website works as a **broker.** Each sensor readings are stored in its own feed. Each feed is then called to be visualized using tools provided by the website. However those feeds can store data, but they can only store for short period of time (few months), so **"Google Sheets"** is used as the cloud database to store this large volume of data permanently and to be accessed from any device where each reading is accompanied with its timestamp providing archive or history over long-term. The system is accompanied with safety alert when the readings of sensors overcome certain threshold which can be indicated by a **buzzer** sound and on the **dashboard.**

Chapter 3: Constructing and testing the prototype

3.1 Materials and Methods

Table 3.1:
Materials

Item	Description	Usage	Cost	Source	Picture
Gas Heater	Fire igniter using natural gas	Heat source for heating container and boiling water	40 L.E.	Gas Cylinders Supplier	
Heating Container	Aluminum 5L Container	Simulate Reactor Core material	60 L.E.	Kitchenware Store	
Water Container	15L Plastic Container	Water Source for Simulation	5 L.E.	Plastics Store	
Hxchen R385 Pump	12V liquids pump	Pump water from container to heater	230 L.E.	Electronics Store	
Potentiometer	three-terminal resistor with a rotating contact	Adjustable voltage divider to control the pump	5 L.E.	Electronics Store	
Hose	120 cm rubber hose of 0.5 cm diameter	Carry Water between container, pump, heater	20 L.E.	Plumbing tools Store	
UV Lamp	A 230V UV Lamp with transformer	Radiate UV light rays	350 L.E.	Electronics Store	
ESP32	Microcontroller	Control Monitoring System	350 L.E.	Electronics Store	
Breadboard	Contact board for creating electronic circuits	Connecting modules to ESP32 without the need of soldering	27 L.E.	Electronics Store	
YF-S401	Water Flow Rate Sensor	Measuring Water Flow Rate from pump to heater	250 L.E.	Electronics Store	
Thermocouple MAX-6675 with k-type sensor	Withstand extreme temperatures	Monitoring Temperature of the heating container	225 L.E.	Electronics Store	
GUVA-S12SD	UV light sensor	Detecting Presence of UV Radiation	120 L.E.	Electronics Store	
Jumper Wires (x15)	Copper wires covered by insulating Plastic	Connecting modules to ESP32	0.75 L.E.	Electronics Store	
Buzzer	Electric to Sound signals converter	Safety Alert	7.5 L.E.	Electronics Store	
Batteries (x5)	3.7V Lithium Ion	Powering the ESP and the Pump	50 L.E.	Electronics Store	
Battery Case Holders (x2)	2 Cells Holder	Connect Batteries in Series	15 L.E.	Electronics Store	
Total Cost		**1,883 L.E.**			

Safety Precautions:

- All construction processes were done in the school fabrication laboratory or under the supervision of an expert person.
- Personal Protective Equipment (PPE) were used as Laboratory coats, safety gloves, safety goggles, and masks.
- Electrical safety was insured using gloves and insulating materials such as wood and wool.

Methods:

First of all, the 120cm hose is cut into three parts of lengths 20cm, 60cm, 40cm for (container to pump), (pump to water flow sensor), (water flow sensor to container) respectively. A hole of diameter 0.5 cm equal to the diameter of the hose is made in the water container using sharp cutter. The 20cm hose is adhered to the hole in the container using wax. The other end of the 20cm hose is inserted into the inlet nozzle of the water pump. From the outlet nozzle of the pump, the 60cm hose is inserted from one of its ends and the other end is inserted in the inlet nozzle of the water flow sensor. For the outlet of the flow sensor, the 40 cm hose is inserted. Finally, all the nozzles of the pump, the flow sensor and even the hole of the container is surrounded by insulating thread seal tape to prevent any water leakage. The water flow sensor is then placed vertically on a piece of wood to pour water into the heating aluminum container placed over the gas heater. Using a heat-resistant tape the thermocouple is attached to the outer surface of the heating container and the UV lamp is also tapped to the wooden column and connected to electricity.

Figure 3.1:
Water Circulation

The second part in the journey of prototype construction is connecting the sensors to the ESP. Two wires are connecting between the 3.3V and GND pins of the ESP and the breadboard to be common power pins for all the components. Each component's GND and VCC pins is then connected by two wires to the GND and 3.3V power pins in the breadboard. The Signal Output (SO), Chip select (CS), and Serial Clock (SCK) pins of the thermocouple are connected to ESP pins D18, D19, D21 respectively. The SO pins of the water flow sensor (yellow wire) and UV light sensor are connected to ESP pins D4 and D25 respectively. The buzzer negative leg is connected to the GND of breadboard while the positive leg connected to ESP D13 pin. The code for controlling the system was written using Arduino IDE, a software app, including functions to connect the ESP to Wi-Fi connection, upload to dashboard, and upload to the google sheets. The code is then uploaded to the ESP32 Board using a Micro USB Data Syncing cable. After the code is uploaded successfully, the ESP is disconnected from the laptop and connected to batteries for power. Two 3.7V batteries are placed in their cells in the battery holder and the positive (red) wire of the holder is connected to the ESP VIN pin and the negative (black) wire is connected to another ESP GND pin. A diagram for the circuit is shown in figure (3.2).

Figure 3.2:
Circuit Diagram

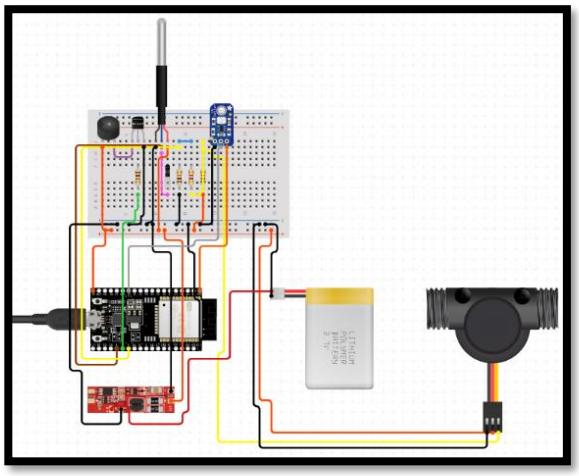
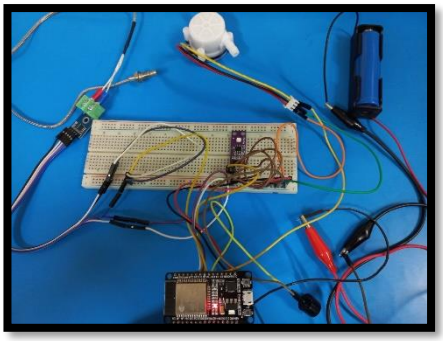
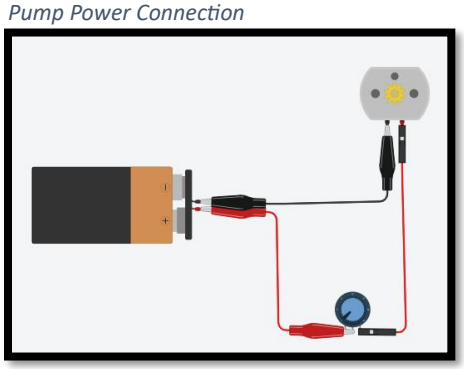


Figure 3.3:
Circuit Connection



For powering the Water Pump, the 12V pump is powered using three 3.7V batteries and to control the speed of output water the power of pump is controlled using a potentiometer. The positive wire of the batteries' holder is connected to the middle leg of potentiometer and one of the lateral legs is connected to positive pole of the pump, while the negative pole of pump is connected directly to the negative wire of the batteries' holder as shown in figure (3.4).

Figure 3.4:
Pump Power Connection



3.2 Test Plan

To ensure our project follows the right path, a three-step test plan was designed to measure the efficiency and integrity of the prototype. Those steps were conducted to test the design requirements, aligning with testing the flow and functionality of the prototype.

The first step was designed to check the presence of UVC radiation. A UV lamp was installed as a UV source. The lamp was firstly turned off to check the sensor reading before UV is present. After readings were recorded, UV lamp was turned on, announcing the presence of a UV light, and sensor reading were recorded again.

Updating the dashboard within a maximum of 10 seconds, as defined in the second design requirement, was tested in the second step. Coolant water flow rate was controlled through controlling the input power to the water pump, using changeable resistance. Then, time between each reading and the next was recorded.

Finally, the emergency alarm that was calibrated to be launched when the core approaches its forging temperature was tested in the third step. The temperature of the core was raised gradually until the forging temperature was reached. Alert responses were recorded four times at four different temperatures.

3.3 Data Collection

When building and testing the prototype, some tools were used to perform measurements, as shown in table 3.2:

Table 3.2:
Measuring Tools

Measuring tools	Measurement	Accuracy	Picture
Vernier caliber	**The diameters of tubes**	**±0.02mm**	
Measuring Ruler	**The length of tubes**	**±0.5mm**	
Thermocouple type k	**Core temperature**	**±3°C**	
YF-S401	**Water flow rate**	**±5%**	
GUVA-S12SD	**Light wavelength**	**±1nm**	

We conducted tests in two different trials, one positive and one negative. The results of each trial were collected and shown below:

Negative Trials:

When GUVA-S12SD UV Sensor detected UVC, the output readings of wavelength weren't in known wavelength range of UVC (100-280 nm), while when not exposed to UV light the readings of wavelength indicated presence of UVC contradicting the design requirements as shown in table (3.3).

Table 3.3:
Negative Trial

UV Lamp State	1st Reading	2nd Reading	3rd Reading	Avg. Wavelength Reading
On	355 nm	366 nm	359 nm	360 nm
Off	246 nm	248 nm	253 nm	249 nm

Positive Trials:

The detected UVC wavelength reading lied in acceptable range achieving the design requirement as shown in table (3.4). Time response of data update on dashboard for any change in monitored factors was less than 5 seconds and the alerting system responded on approaching forging temperature achieving all design requirements as shown in tables (3.5), (3.6).

Table 3.4:
Positive Trial of UVC Detection

UV Lamp State	1st Reading	2nd Reading	3rd Reading	Avg. Wavelength Reading
On	265 nm	252 nm	248 nm	255 nm
Off	352 nm	359 nm	357 nm	356 nm

Table 3.5:
Time Response of Data Update on Dashboard

Trials	Operating Voltage	Water Flow Rate	Time Response
1st Trial	6V	2.5 L/min	8 secs
2nd Trial	9V	4.3 L/min	9 secs
3rd Trial	12V	5.8 L/min	8 secs
Average Time Response = 8.3 secs			

Table 3.6:
Alert Response to Temperature Increase

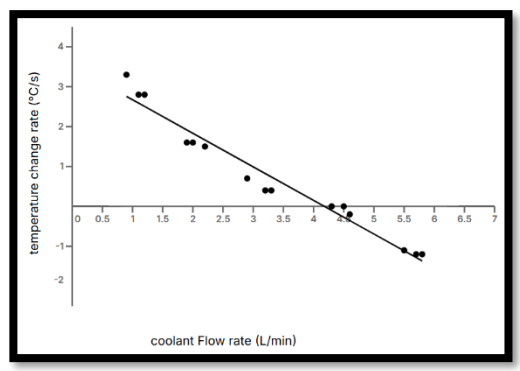
Core Temperature Reading	120 °C	184 °C	213 °C	227 °C	231 °C
Alert Response	Off	Off	Off	On	On

4.1 Analysis and Discussion

Improving Egypt's scientific and technological environment is a critical issue that affects the country's overall development in all sectors. In particular, the lack of modern methods for monitoring energy production is a significant challenge facing the energy sector. As Egypt turned to Nuclear Power as a modern energy source, enhancing the efficiency and safety of nuclear plants is a crucial concern. A significant challenge in nuclear plants is the extreme dependency on human power, which threatens the plant's efficiency and safety. To address this problem, an automated monitoring system is needed.

The system will track the critical parameters that hold the plant's production efficiency and ensure human safety. Those parameters are nuclear core temperature, coolant water flow rate, and UV excess radiations. First, monitoring the temperature of the nuclear reactor is crucial as it directly controls the rate of water turning to steam in advance of driving the turbines, determining the reactor's overall efficiency. In addition, when the core temperature approaches its limit, the forging temperature, indicates the possibility of a disaster. Next, the coolant water flow rate should be closely traced to ensure the core temperature doesn't exceed the threshold safety value, avoiding sudden eruptions in the nuclear core. Coolant water affects the core temperature directly as water flow rate is inversely correlated with the rate of change of temperature, the mor amounts of water, the more the heat absorption, and the lower the core temperature. Using scatter plotting, a line of regression was drawn in the form of Y= mX + b. to predict the rate of the change from the flowrate, as shown in figure (4.1). The slope(m) and y-intercept(b) of the regression line was calculated to be -0.84 and 3.5 respectively using the following equation:

Figure 4.1:
Correlation between water flow rate and change

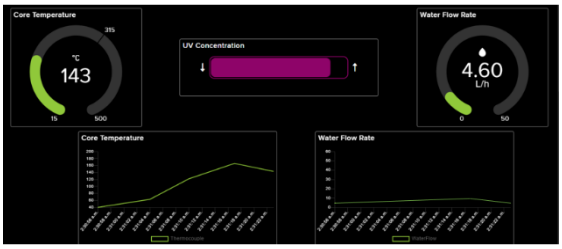


$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \quad , \quad b = \frac{\sum y - m\sum x}{n}$$

(Cook & Upton, 1996)

Further, excess UV radiation causes extreme human fatigue and indicates system deficiencies; so, detecting radiation leakage is essential in nuclear plants. Finally, visualizing the collected data is the cornerstone for taking control of the monitoring process and enhancing monitoring efficiency, allowing authorities to take a life overview of the plant.

Figure 4.2:
Data Visualization



For this purpose, electronic sensors were used to detect each parameter minimizing the likelihood of human errors, while IoT techniques were used for data visualization as shown in figure (4.2), including using Wi-Fi protocols for wireless data transfer. Google spreadsheets, were used for data storage due to its cloud-based capabilities which enhance data accessibility, storing, and updating in real-time. When it comes to sensors, the thermocouple temperature module K-type chosen rather than S, N, and R types, the K type is the only module detecting temperatures within our desired range, 0 °C and 800 °C. Regarding water flow rate, water flow module YF-S401 was the best choice due to its flow rate range, 1-30 L/min, aligning with its accuracy (±10), in addition to its efficient mechanism, which is driven by moment of inertia about the center. For UV, the GUVA-S12SD UV sensor module was preferred over the GY-8511, because its detection range of 240 nm to 370 nm includes UVC radiation, which is the most prevalent type produced in nuclear plants. To enhance the durability of the system, lithium-ion batteries are used as a power source, with a lifespan of minimum 6 months. To choose appropriate batteries, the total capacity needed by the system was calculated through the law C = Power × time. With a voltage of 3.3V, and total consumed current(I) of 271 mA, the consumed Power was calculated to be 0.89-Watt, P = I × V; so, the total capacity needed is calculated to be 3863.4 Watt-hours.

Following the construction of the prototype, the UVC presence, updating the dashboard within a maximum of 10 seconds, and the alert for the forging temperature were tested. UV lamp was used to test the detection of UVC. At first, irrational readings were present. When the UV light was turned off, the readings indicated a wavelength of 240 instead of 370, which indicates UVC presence. Conversely, when the lamp was turned on, a wavelength of 370 was indicated instead of 240, which indicates UVC absence. This issue stemmed from a technical issue in the Arduino code, specifically with the use of the map() function. The map function in Arduino is used to re-map a variable from one range to another. It takes

five values: the variable to be mapped, the 2 limits of the original range, and the 2 limits of the new desired range. The problem arose from a misunderstanding of the UV wavelength ranges. In the light spectrum, as the frequency (F) increase, the wavelength (λ) decrease, according to the law:

$$\mathbf{C = \lambda F}$$

where C is the speed of light (Serway et al., 2016). The UV light ranges between 100 to 370 nanometers (nm) **(**further divided into UVA (315-400 nm), UVB (315-280 nm), and UVC (280-100 nm) (Ultraviolet (UV) Radiation, 2017). We attempted to convert the UV sensor's electrical pulse readings, which range from 0 to 4095, to a corresponding wavelength. However, mistakenly reversed the output range, mapping the input to an output range of 240-370 nm instead of 370-240, which resulted in indicating UVC presence when the lamp was off. After that, the dashboard update was tested to ensure the design requirement is achieved, as data should be updated within a maximum of 10 seconds. The power supplied to the water pump was controlled, adjusting the water flow rate. Consequently, the water flow rate sensor reading was changed, and the dashboard was updated successfully within 8 seconds as shown in figure (4.3). Finally, the temperature was adjusted simultaneously to test the alert. Alert was launched on approaching the forging temperature at 227°C with a margin of 23°C. The second trial was positive, after considering coding issues and running the modified version of the code, with UV readings of nearly 255 nm.

Figure 4.3:
The timestamp of each reading

1	Nuclear Station - Data Storgae			
2	Time Stamp	UV	Temperature	Flow Rate
3	02/12/2024 11:37:46	245	65	0.9
4	02/12/2024 11:37:54	243	92	1.1
5	02/12/2024 11:38:01	256	117	1.2
6	02/12/2024 11:38:08	262	130	1.9
7	02/12/2024 11:39:07	248	143	2.2
8	02/12/2024 11:39:15	263	152	2
9	02/12/2024 11:39:23	258	155	3.3
10	02/12/2024 11:39:29	255	159	2.9
11	02/12/2024 11:39:37	263	162	3.2
12	02/12/2024 11:39:43	265	163	4.3
13	02/12/2024 11:39:50	272	163	4.6
14	02/12/2024 11:39:57	264	161	4.5
15	02/12/2024 11:40:04	262	157	5.5

4.2 Recommendations

Finally, the project has come to an end. Here are some recommendations to improve this project and ensure that it works more efficiently in the future. These recommendations are as follows:

• Monitoring other types of radiation, like alpha, beta, and gamma, to enhance protection against all kinds of leaks.

• Applying machine learning and Artificial Intelligence to radiation sensors reading to distinguish true radiation signals from noise and to cancel the noise more efficiently using destructive interference.

• Using optical fibers to connect the sensors and microcontroller, as they offer speeds 10 to 20 times faster than normal cables, enhancing the Real-Time presentation of the data.

• Coating the sensors with radiation absorbing materials like lead to enhance accuracy in high radiation zones.

• Developing and designing a website with custom-built data visualization features and user interface, instead of pre-made online visualization tools.

• Implementing the monitoring system in the upcoming Dabaa Powerplant

4.3 Learning Outcomes

Table 4.1:
Learning Outcomes

Subject	Learning Outcome	Application
Biology	BI.3.01	Electronic sensors receive and transmit data, translating it into electrical pulses, like the human nervous system, which receives and transmits the signals from human sensations.
Chemistry	CH.1.13	Exothermic reactions are the main part of the nuclear core in our project. As nuclear fission happens, it releases massive amounts of energy, in the form of heat, that further changes into electrical energy.
Computer Science	CS.1.05	Google spreadsheets were used for data storage due to its cloud-based capabilities which enhance data accessibility, storing, and updating in real-time.
Earth Science	ES.3.02	GPS consists of a transmitter, a transducer, and a receiver. The same is the electronic sensors, which receive information, translate it, and then send it in the form of electrical pulses.
English	ENW. 3.1.02	Crafting an argumentative essay preferring between different modules of the sensors requires critical analysis and persuasive rhetoric to advocate for the most effective architectural choices.
Mechanics	ME.3.01	The mechanism in which the water flow sensor captures its readings relies on momentum. The passing water acts as the force that affects the sensor's blades causing it to rotate. This rotation produces proportional electrical pulses.
Physics	PH.3.02	Studying waves and their properties and parameters helped us understand the principle of the UV sensor, which works by receiving different rays and translating the waves in the UV wavelength range, which is 370-240 nm.
	PH.3.05	The study of different communication methods and their details helped us choose Wi-Fi as our method of transmitting data, for its high speed and availability.
Robotics	RAL.3.07	We compared DC and AC motors and chose DC as they are compatible with the incoming current from the battery system.
Statistics	ST.3.02	The correlation was found between flow rate and rate of change of the core temperature; so, using a scatter plot and regression line enabled us to predict the nature of the rate of change of temperature.

4.4 References

Are Egypt's manufacturing industries embracing the shift to industrial automation? (2022, April 10). Enterprise - The State of Nation. https://enterprise.press/industries/egypts-manufacturing-industries-embracing-shift-industrial-automation/#:~:text=Multinationals%20are%20leading%20the%20automation,medium%20enterprises%20(MSMEs)%20lack%20the

Bohigas, X., & De Fortuny, T. (2014, March 14). *The incidence of human error in the handling of nuclear weapons*. Centre of Studies for Peace J.M. Delàs. https://centredelas.org/actualitat/the-incidence-of-human-error-in-the-handling-of-nuclear-weapons/?lang=en

Bohigas, X., & De Fortuny, T. (2024, March 3). *The incidence of human error in the handling of nuclear weapons*. Centre of Studies for Peace J.M. Delàs. https://centredelas.org/actualitat/the-incidence-of-human-error-in-the-handling-of-nuclear-weapons/?lang=en

Cook, I., & Upton, G. (1996). Regression and Correlation. Oxford University Press.

Darwish, I. I. (2022). *University-Industry-Collaborations in Egypt: Academics' Perception of Motivators and Success Factors.* The American University in Cairo (AUC), Public Policy & Administration Department. AUC Knowledge Fountain . https://fount.aucegypt.edu/etds/1903/

Green, C. (2023, August 22). *The Transformational Role of IIoT in Asset Management*. IIot World. https://www.iiot-world.com/industrial-iot/connected-industry/the-transformational-role-of-iiot-in-asset-management/#:~:text=A%20report%20by%20the%20World%20Economic%20Forum%20indicates%20that%20IIoT,operating%20expenses%20for%20manufacturing%20companies.

Media Center. (2022, December 28). State Information Service. https://sis.gov.eg/Story/174654/Egypt-ranked-35th-globally-in-terms-of-spending-on-R%26D?lang=en-us

Research and development expenditure (% of GDP). (2024, April 24). World Bank Group. https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS?locations=EG

Rostedt, J., & Koivunen, L. (2021). *Mapping of Condition Monitoring Systems in Olkiluoto Nuclear Powerplants p 4, 33-49.* Energiforsk.

Saad, E. M. (2015). *Nuclear Reactors p 18.* Fayoum: Faculty of Engineering Fayoum University.

Serway, R. A., Vuille, C., & Hughes, J. (2016). Vibrations ans Waves. In *College Physics* (11th ed., Vol. 1, pp. 444, 445). Cengage.

Teollisuuden Voima Oyj. (2024, January 3). *Olkiluoto produced last year about 31 percent of the electricity consumed in Finland* . TVO. https://www.tvo.fi/en/index/news/pressreleasesstockexchangereleases/2024/4718828.html

THE 17 GOALS. (2024, April 20). United Nations. https://sdgs.un.org/goals

Ultraviolet (UV) Radiation. (2024, November 16). (University Corporation for Atmospheric Research) Center for Science Education. https://scied.ucar.edu/learning-zone/atmosphere/ultraviolet-uv-radiation

World Nuclear Association. (2024, April 29). *Fukushima Daiichi Accident*. World Nuclear Association. https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/fukushima-daiichi-accident